

Wanglong Lu

Senior Data Scientist | ML Systems & AI Engineering

St. John's, NL, Canada | lwlxhl@gmail.com | Website | GitHub | LinkedIn | Google Scholar

Profile

Computer Science Ph.D.; **30 scholarly works overall** (author-reported). Builds **AWS-backed ML tools and AI applications**, including **SageMaker job submission** and an **internally used RAG assistant**; develops models in PyTorch.

Technical Skills

Engineering: Python, SQL, Git, Bash, C++, Docker; FastAPI, Mangum, PostgreSQL.

Cloud & Compute: AWS SageMaker, Lambda, Bedrock, S3; PyTorch, CUDA, SLURM.

ML & GenAI: Feature engineering, Transformers, diffusion, PEFT/LoRA, Hugging Face, RAG, agentic tools.

Professional Experience

Nasdaq

St. John's, NL, Canada

Senior Data Scientist, AI/Analytics

Mar. 2025 – Present

- Delivered **MLP application integration** into a **SageMaker workspace** from an existing cross-team UI/backend prototype; configured **networking and IAM**. Platform at **internal deployment stage**.
- Implemented **PostgreSQL-backed user-state persistence**; deployed **FastAPI services on AWS Lambda** using **Mangum** and connected them to the platform application layer.
- Built **SageMaker Job Submitter for MLP**: script/dependency packaging for **Processing, Training, HPO and Pipelines**; cut **new-task configuration/submission preparation from 7 days to 1 day (about 86%)**.
- Developed **PagerDuty RAG assistant v2** for incident-context retrieval and troubleshooting guidance; **now used internally and recognized in a company shout-out**.
- Designed and implemented **feature engineering with PyTorch/CUDA-accelerated feature-label AUROC screening**; used **Amazon Bedrock** to assess candidates' semantic plausibility.
- Designed cross-scale ensemble feature selection for **cheque-fraud ML (CFML)**, removing **77.2% of input features** while retaining competitive performance in evaluations.
- Mentored projects in **input payload checking, GNN-based crime-ring detection, cheque OCR and tabular-data generation**; the latter led to a **StDDPM manuscript submission**.

AI Algorithm Intern (two placements)

May 2024 – Jan. 2025

- Designed and implemented **TextDoctor** in **Python/PyTorch** for document/cheque inpainting; ran AWS/SLURM/CUDA experiments and demonstrated **8K inpainting on a 24GB RTX 3090**.
- Evaluated **7 document datasets** without dataset-specific fine-tuning. FUNSD PaddleOCR word accuracy: **55.38% vs. DocDiff's 45.24% (+10.14 percentage points)** using TextDoctor (GSDM).
- **First-author TextDoctor manuscript submitted to EAAI**; the publicly linked version is an **arXiv preprint**, not an accepted journal article.
- Presented research after **both internship placements**: about **40–60 attendees** at the earlier talk and **39 cross-functional attendees** at the final talk.

Beaufort Solutions

St. John's, NL, Canada

AI Intern

Jul. 2022 – Nov. 2022

- Led design of **TEG**, a **CLIP-based few-shot adaptation** method for photo-book curation, using a text-embedding-guided classifier and auxiliary loss; evaluated with Python/PyTorch and AWS/SLURM.
- Built the collection pipeline and led **Theme25 dataset construction: 35,655 images, 25 themes**; evaluated three datasets; research published in **Journal of Electronic Imaging (2024)**.

Zhongkong Hanlian

Hangzhou, China

AI Algorithm Intern

Dec. 2017 – Aug. 2018

- Deployed **vehicle-logo recognition with Caffe** at **community security gates**; designed **ResNet–DenseNet fusion**, built Java annotation tools, and constructed a training/evaluation dataset.

Agentic AI Tooling

Local Agentic Code Editor

Nasdaq internal exploratory prototype; 2026

- Built a working local code editor connecting **language-model APIs** to **task-dependent routing across code generation, code review and shell execution**.
- Implemented an iterative debugging loop: **diagnose bugs, edit code, apply fixes and inspect execution feedback** to determine the next action. Remains an exploratory prototype, not an approved production product.

Selected Model Development & Publications

UltraDiffEdit – High-Resolution Generative Editing [Code]

First author; IEEE TNNLS 2026

- Extended pretrained diffusion models to **tuning-free 8K editing on one RTX 3090** with multi-patch encoding and global-local sampling; tested SDXL/ControlNet on **three benchmarks (1,188 images)**.
- **UHRSEdit (4K–8K) crop-FID 7.45 vs. 18.76 for SDXL+bicubic (60.3% lower)**.

AdaMSS – Parameter-Efficient Adaptation [Code]

Second author; NeurIPS 2025

- Co-developed multi-subspace tuning; **ViT-Large accuracy 90.13% vs. LoRA's 85.40%** across **7 tasks** with **15.4% of LoRA's trainable parameters**. Method integrated into Hugging Face PEFT.

DocPure – Unified Document Restoration [Code]

Co-first author; IEEE TCSVT 2026

- Co-developed prompt-free, structure-guided wavelet restoration for **4 tasks and 12 datasets**. On TDD deblurring vs. DocRes: **PSNR 32.55 vs. 28.34 dB; PaddleOCR word accuracy 61.27% vs. 55.66%**.

Visual Style Prompt Learning [Code]

First author; Pattern Recognition 2025

- Developed diffusion style prompts: **four denoising steps, 12.3 FPS at 512 × 512 on RTX 3080 Ti; WebPhoto-Test FID 74.25 vs. CodeFormer's 83.17 (10.7% lower)**.
- On **CelebA-Test**, FAN landmark NME **6.08% to 2.43%**; HSEmotion agreement **79.73% to 86.03%**, using the respective models' predictions on clean images as references.

FACEMUG – Multimodal Local Facial Editing

First author; TVCG 2025 (online 2024)

- Designed feature fusion and latent warping for **six input modalities; 77.88% preference over Unite&Conquer** in a **52-participant CelebA-HQ study** with sketch/semantic/text guidance.

GRIG – Data-Efficient Image Inpainting [Code]

First author; CVM 2025

- Author-manuscript evaluation of CNN–Transformer residual inpainting: **10 datasets (18–750 training images); 3.70–26.48% lower FID vs. next-best baselines at 50–60% masking**.

Full publication record: Google Scholar and publication portfolio.

Education

Ph.D., Computer Science – Memorial University of Newfoundland, Canada

2021–2025

M.Eng., Computer Software and Theory – Wenzhou University, China

2018–2021

B.Eng., Digital Media Technology – Communication University of Zhejiang, China

2014–2018

Selected Recognition & Service

Outstanding Research Award (Ph.D., Memorial, 2023). **Five granted Chinese invention patents as co-inventor**; assignee at grant: Wenzhou University.

Mentored **12 students** in generative vision and applied ML, including restoration and recognition projects.

Teaching assistant in Data Structures and Algorithms (2021) and Introduction to Machine Learning (2024).

Invited reviewer for **IEEE TPAMI, TIP, TMM, TCSVT, JBHI and SPL; Pattern Recognition, EAAI, JVCi, Knowledge-Based Systems, Neurocomputing, Applied Soft Computing, Scientific Reports, Displays; ECCV 2026**.

Presented feature-selection research to **60+ attendees** (2026); invited speaker, IEEE Newfoundland and Labrador Section AGM (2026); guest lecturer on deep generative models, Memorial University (2025).